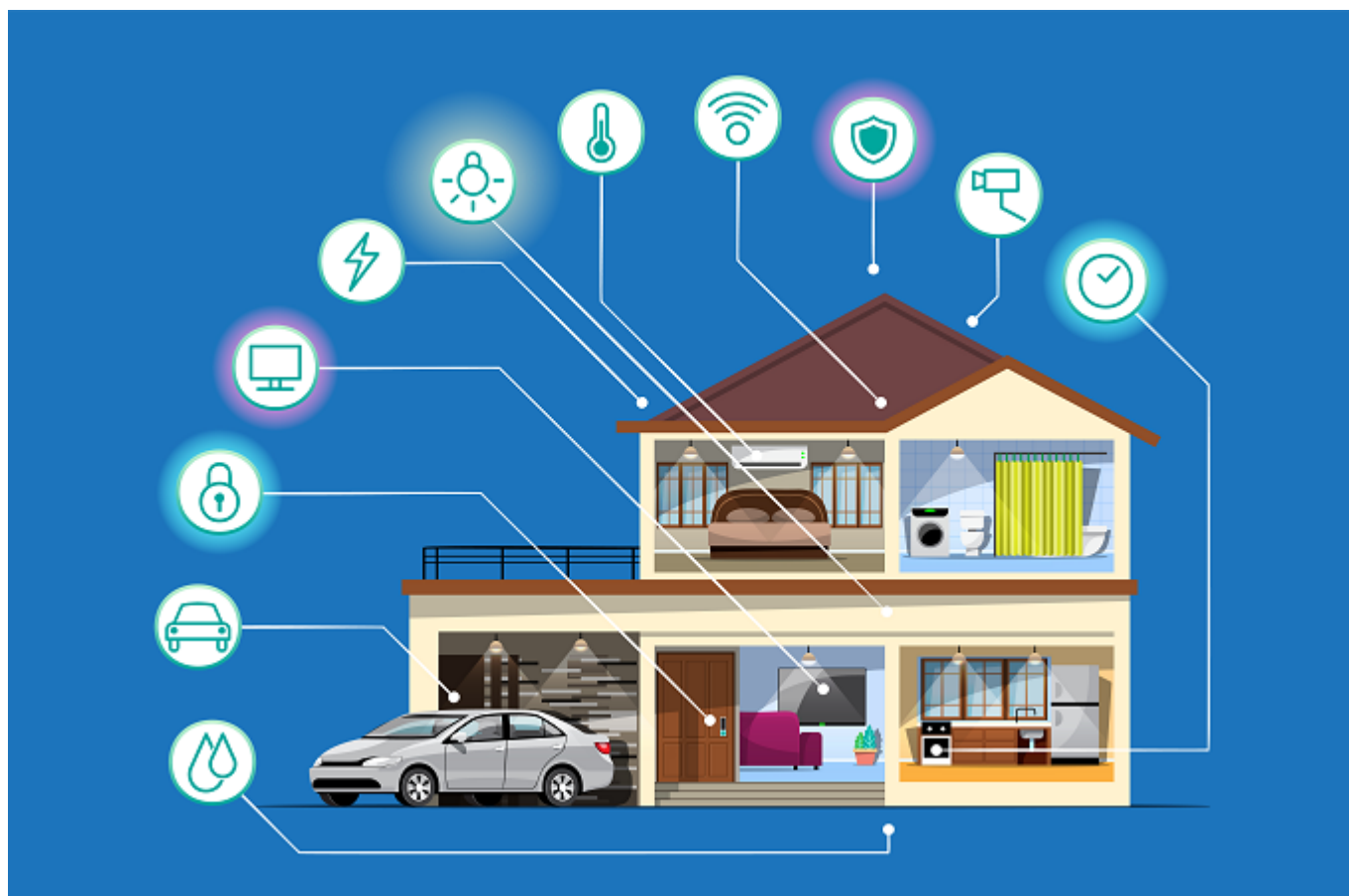


## Qualification Pack



# Building Automation Specialist

QP Code: IAS/Q3006

Version: 6.0

NSQF Level: 5

## Qualification Pack

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## Qualification Pack

### IAS/Q3006: Building Automation Specialist

#### Brief Job Description

Performs system design, wiring, integration, testing, installation & maintenance of automation systems in buildings involving HVAC, FAS, ACS & CCTV Systems.

#### Personal Attributes

This job requires interdisciplinary aptitude, ability to learn, ability to deal with a variety of technology and people of different trades and skills.

#### Applicable National Occupational Standards (NOS)

##### Compulsory NOS:

1. [IAS/N2103: Install and Provide Technical Support for CCTV Surveillance Systems](#)
2. [IAS/N2104: Integrating and Controlling Building automation System](#)
3. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)
4. [IAS/N8014: Design, install, & Provide Technical Support for the HVAC System](#)
5. [IAS/N8015: Design Install and Provide Technical Support for Fire Alarm System](#)
6. [IAS/N8016: Install and Provide Technical Support for Access Control Systems](#)
7. [IAS/N8012: Integration of Smart Technologies in HVAC, Lighting, Security and other building Automation](#)

#### Qualification Pack (QP) Parameters

|                   |                                   |
|-------------------|-----------------------------------|
| <b>Sector</b>     | Instrumentation                   |
| <b>Sub-Sector</b> | Instrumentation & Automation      |
| <b>Occupation</b> | Product Engineering/System Design |
| <b>Country</b>    | India                             |
| <b>NSQF Level</b> | 5                                 |
| <b>Credits</b>    | 19                                |

## Qualification Pack

|                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Aligned to NCO/ISCO/ISIC Code</b>                      | NCO-2015/2149.0100                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Minimum Educational Qualification &amp; Experience</b> | <p>Completed 3 year diploma after 10th (Engg Diploma Instrumentation/Mechanical/EEE/ECE/Mechtronics) with 1.5 years of experience</p> <p>OR</p> <p>B.E./B.Tech (2nd Year Completed Instrumentation/Mechanical/EEE/ECE/Mechatronics)</p> <p>OR</p> <p>Previous relevant Qualification of NSQF Level (4.5) with 1.5 years of experience Building Automation Domain</p> <p>OR</p> <p>Previous relevant Qualification of NSQF Level (4) with 3 Years of experience Building Automation Domain</p> |
| <b>Minimum Level of Education for Training in School</b>  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Pre-Requisite License or Training</b>                  | NA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Minimum Job Entry Age</b>                              | 18 Years                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Last Reviewed On</b>                                   | NA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Next Review Date</b>                                   | 07/10/2030                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>NSQC Approval Date</b>                                 | 07/10/2025                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Version</b>                                            | 6.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Reference code on NQR</b>                              | QG-05-CG-045542025-V2-IASC                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>NQR Version</b>                                        | 2.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

## Qualification Pack

# IAS/N2103: Install and Provide Technical Support for CCTV Surveillance Systems

## Description

This OS unit is about Installing and Providing Technical Support for CCTV Surveillance Systems at the customers premises

## Scope

The scope covers the following :

- This unit/task covers the following: Capturing the requirements of CCTV Surveillance Systems by site survey Suggesting and taking approval from the customer for CCTV System to be installed Installing approved CCTV components as per site requirements Wiring Electrical and Electronics components as per specifications Testing CCTV systems at customer premises Providing Technical Support for CCTV Systems Achieving Quality and Productivity as per company norms

## Elements and Performance Criteria

### *Capturing the requirements of CCTV Surveillance Systems by site survey*

To be competent, the user/individual on the job must be able to:

- PC1.** Capturing work requirements of the client by site survey
- PC2.** Developing BOQ according to the requirement of the client
- PC3.** Assisting customer about different types of technologies used in CCTV Surveillance Systems according to the need of site
- PC4.** Ensuring that Suggesting components matches to customers requirement
- PC5.** Assisting the customers about the company policies towards services and warranty
- PC6.** Managing proper documentation of site survey and customer requirements

### *Suggesting and taking approval from the customer for CCTV System to be installed*

To be competent, the user/individual on the job must be able to:

- PC7.** Suggesting appropriate CCTV components to the customer according to the site
- PC8.** Assisting the customers about technologies used in CCTV systems with their specifications
- PC9.** Taking approval for installing CCTV Systems from the customer
- PC10.** Preparing and assembling CCTV Systems components as per the requirement
- PC11.** Creating check list before going to the site for installation
- PC12.** Maintaining complete documentation of the components to be installed

### *Installing approved CCTV components as per site requirements*

To be competent, the user/individual on the job must be able to:

- PC13.** Assisting procurement of hardware required for CCTV system installation
- PC14.** Collecting and checking of components before going to customer premises
- PC15.** Assisting technicians for checking hardware components before Installation
- PC16.** Replacing components if found malfunctioning

## Qualification Pack

- PC17.** Preparing Checklist and ensure the availability of every component before installation
- PC18.** Installing CCTV components at the customers site
- PC19.** Mounting the CCTV camera so as to cover maximum area
- PC20.** Deciding on the height of camera installation according to the end purpose
- PC21.** Setting up the camera such as pan, tilt, zoom unit as per customer requirements
- PC22.** Set camera controls
- PC23.** Connecting the power and video output cable to the camera
- PC24.** Ensuring that all the hardware matches the customer requirement, agreed features and specifications

### *Wiring Electrical and Electronics components as per specifications*

To be competent, the user/individual on the job must be able to:

- PC25.** Determining the type of cable requirement for different types of network type such as USB, twisted pair, etc.
- PC26.** Ensuring adequate length of cables are available for installation
- PC27.** Wiring Power Supplies, Earthing & Grounding.
- PC28.** Laying the cables in the building or site to connect the camera and system
- PC29.** Using BNC connectors for joining cables and crimping them
- PC30.** Using power cable of specified thickness to connect CCTV system with power supply
- PC31.** Connecting all the cables from multiple cameras to the CCTV system area

### *Testing CCTV Components at customer premises*

To be competent, the user/individual on the job must be able to:

- PC32.** Checking voltage and resistance at all appropriate points of the system
- PC33.** Correcting alignment and operation of CCTV hardware
- PC34.** Checking correct operation of each component. Ensure that there are no malfunctioning, if yes then replace the component
- PC35.** Checking the signals of CCTV components
- PC36.** Checking all the data received by CCTV as per the requirement
- PC37.** Ensuring that there are no cable joins, sharp bends during cabling
- PC38.** Ensure 100% satisfaction from customers for all the installed components

### *Providing Technical Support for CCTV Systems*

To be competent, the user/individual on the job must be able to:

- PC39.** Providing Technical Support for CCTV devices at the customer premises
- PC40.** Commissioning CCTV Systems performance as per customer requirements
- PC41.** Achieving zero errors in commissioning as per company policy
- PC42.** Identifying problems and alert on time
- PC43.** Fixing for any errors (if any) identified
- PC44.** Verifying software implementation checks;(AI, AO, DI,DO) I/O points
- PC45.** Ensuring zero-material damage while handling the equipment during installation process
- PC46.** Performing software functionality test

### *Achieving Quality and Productivity as per company norms*

To be competent, the user/individual on the job must be able to:

## Qualification Pack

- PC47.** Achieving 100% work schedule as planned for the week
- PC48.** Following standard operating procedure of tools and equipment and avoid any hazard
- PC49.** Achieving zero component damage
- PC50.** Keeping work area clean and organized
- PC51.** Achieving 100% compliance with health and safety guidelines and rules
- PC52.** Ensuring installed components as per company norms and standards

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Company's policies on: incentives, delivery standards and personnel management, customer management
- KU2.** Reporting and documentation processes
- KU3.** Importance of the individual's role in the workflow
- KU4.** Reporting structure
- KU5.** Complete knowledge of CCTV Surveillance System
- KU6.** Knowledge of Ethernet and I/P & Industrial Networks
- KU7.** Operation of components used in CCTV system
- KU8.** Elements of CCTV systems such as camera, DVR, monitor
- KU9.** Electrical/electronic/mechanical parts required for the ACS
- KU10.** Material requirement of components with their specifications
- KU11.** Installation procedures given in the manuals
- KU12.** Different type of cables used for data transmission and power transmission
- KU13.** Power requirement of different CCTV related equipment
- KU14.** Knowledge of communication between CCTV and other BAS components
- KU15.** Video recording of footage analog and digital
- KU16.** Different types of camera available in the market
- KU17.** Camera specifications such as focus, lens type, zoom
- KU18.** Controls of different options in camera such as rotation, speed of movement in pan / tilt camera
- KU19.** Voltage and power requirement for different hardware devices
- KU20.** How to operate the system and other hardware
- KU21.** Understanding of Networks protocols
- KU22.** Safety rules, policies and procedures
- KU23.** Quality standards to be followed

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Preparing indents and list of equipment required for the specified Access Controls System

## Qualification Pack

- GS2.** Compiling all the related documents as per the requirements
- GS3.** Record Faults in instruments and submit to the team leader.
- GS4.** Read warnings, instructions and other text material on product levels, components etc.
- GS5.** Reading drawings and job sheets or work orders
- GS6.** Discuss work load with superiors
- GS7.** Communicate with team leader for proper understanding of work
- GS8.** Interact with coworkers and gather all the information related to process requirement
- GS9.** Make decisions according to the situations
- GS10.** Plan the execution of entire design activity; long term and short term activities so that he can finish the task activity wise in the stipulated time.
- GS11.** Organize expert support from/within outside professional environment for new technology and feasibility studies.
- GS12.** Understand real needs of the customer and suggest most appropriate solution
- GS13.** Support customer when they need help
- GS14.** Think and provide best possible solution required for the system
- GS15.** Identify immediate or temporary solutions to resolve delays
- GS16.** Use the existing information to arrive at actionable decision points
- GS17.** Use the existing information for improving the customer satisfaction
- GS18.** Use the existing information to optimize solution and company business
- GS19.** Analyze problems and identify causes and possible solutions
- GS20.** Analyze the work flow and suggest superiors to provide improvement for better production



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                     | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Capturing the requirements of CCTV Surveillance Systems by site survey</i>                                                        | <b>7</b>     | <b>9</b>        | -             | -          |
| <b>PC1.</b> Capturing work requirements of the client by site survey                                                                 | 2            | 1               | -             | -          |
| <b>PC2.</b> Developing BOQ according to the requirement of the client                                                                | 1            | 2               | -             | -          |
| <b>PC3.</b> Assisting customer about different types of technologies used in CCTV Surveillance Systems according to the need of site | 1            | 2               | -             | -          |
| <b>PC4.</b> Ensuring that Suggesting components matches to customers requirement                                                     | 1            | 2               | -             | -          |
| <b>PC5.</b> Assisting the customers about the company policies towards services and warranty                                         | 1            | 1               | -             | -          |
| <b>PC6.</b> Managing proper documentation of site survey and customer requirements                                                   | 1            | 1               | -             | -          |
| <i>Suggesting and taking approval from the customer for CCTV System to be installed</i>                                              | <b>6</b>     | <b>8</b>        | -             | -          |
| <b>PC7.</b> Suggesting appropriate CCTV components to the customer according to the site                                             | 1            | 2               | -             | -          |
| <b>PC8.</b> Assisting the customers about technologies used in CCTV systems with their specifications                                | 1            | 1               | -             | -          |
| <b>PC9.</b> Taking approval for installing CCTV Systems from the customer                                                            | 1            | 1               | -             | -          |
| <b>PC10.</b> Preparing and assembling CCTV Systems components as per the requirement                                                 | 1            | 1               | -             | -          |
| <b>PC11.</b> Creating check list before going to the site for installation                                                           | 1            | 1               | -             | -          |
| <b>PC12.</b> Maintaining complete documentation of the components to be installed                                                    | 1            | 2               | -             | -          |
| <i>Installing approved CCTV components as per site requirements</i>                                                                  | <b>12</b>    | <b>19</b>       | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                           | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC13.</b> Assisting procurement of hardware required for CCTV system installation                                       | 1            | 1               | -             | -          |
| <b>PC14.</b> Collecting and checking of components before going to customer premises                                       | 1            | 1               | -             | -          |
| <b>PC15.</b> Assisting technicians for checking hardware components before Installation                                    | -            | -               | -             | -          |
| <b>PC16.</b> Replacing components if found malfunctioning                                                                  | 1            | 1               | -             | -          |
| <b>PC17.</b> Preparing Checklist and ensure the availability of every component before installation                        | 1            | 3               | -             | -          |
| <b>PC18.</b> Installing CCTV components at the customers site                                                              | 2            | 3               | -             | -          |
| <b>PC19.</b> Mounting the CCTV camera so as to cover maximum area                                                          | 1            | 2               | -             | -          |
| <b>PC20.</b> Deciding on the height of camera installation according to the end purpose                                    | 1            | 2               | -             | -          |
| <b>PC21.</b> Setting up the camera such as pan, tilt, zoom unit as per customer requirements                               | 1            | 2               | -             | -          |
| <b>PC22.</b> Set camera controls                                                                                           | 1            | 2               | -             | -          |
| <b>PC23.</b> Connecting the power and video output cable to the camera                                                     | 1            | 1               | -             | -          |
| <b>PC24.</b> Ensuring that all the hardware matches the customer requirement, agreed features and specifications           | 1            | 1               | -             | -          |
| <i>Wiring Electrical and Electronics components as per specifications</i>                                                  | <b>9</b>     | <b>16</b>       | -             | -          |
| <b>PC25.</b> Determining the type of cable requirement for different types of network type such as USB, twisted pair, etc. | 1            | 3               | -             | -          |
| <b>PC26.</b> Ensuring adequate length of cables are available for installation                                             | 2            | 3               | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                                      | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC27.</b> Wiring Power Supplies, Earthing & Grounding.                                                                             | 2            | 3               | -             | -          |
| <b>PC28.</b> Laying the cables in the building or site to connect the camera and system                                               | 1            | 2               | -             | -          |
| <b>PC29.</b> Using BNC connectors for joining cables and crimping them                                                                | 1            | 2               | -             | -          |
| <b>PC30.</b> Using power cable of specified thickness to connect CCTV system with power supply                                        | 1            | 1               | -             | -          |
| <b>PC31.</b> Connecting all the cables from multiple cameras to the CCTV system area                                                  | 1            | 2               | -             | -          |
| <i>Testing CCTV Components at customer premises</i>                                                                                   | <b>7</b>     | <b>15</b>       | -             | -          |
| <b>PC32.</b> Checking voltage and resistance at all appropriate points of the system                                                  | 1            | 2               | -             | -          |
| <b>PC33.</b> Correcting alignment and operation of CCTV hardware                                                                      | 1            | 2               | -             | -          |
| <b>PC34.</b> Checking correct operation of each component. Ensure that there are no malfunctioning, if yes then replace the component | 1            | 3               | -             | -          |
| <b>PC35.</b> Checking the signals of CCTV components                                                                                  | 1            | 2               | -             | -          |
| <b>PC36.</b> Checking all the data received by CCTV as per the requirement                                                            | 1            | 2               | -             | -          |
| <b>PC37.</b> Ensuring that there are no cable joins, sharp bends during cabling                                                       | 1            | 2               | -             | -          |
| <b>PC38.</b> Ensure 100% satisfaction from customers for all the installed components                                                 | 1            | 2               | -             | -          |
| <i>Providing Technical Support for CCTV Systems</i>                                                                                   | <b>8</b>     | <b>16</b>       | -             | -          |
| <b>PC39.</b> Providing Technical Support for CCTV devices at the customer premises                                                    | 1            | 2               | -             | -          |
| <b>PC40.</b> Commissioning CCTV Systems performance as per customer requirements                                                      | 1            | 2               | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                    | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC41.</b> Achieving zero errors in commissioning as per company policy                           | 1            | 2               | -             | -          |
| <b>PC42.</b> Identifying problems and alert on time                                                 | 1            | 2               | -             | -          |
| <b>PC43.</b> Fixing for any errors (if any) identified                                              | 1            | 2               | -             | -          |
| <b>PC44.</b> Verifying software implementation checks;(AI, AO, DI,DO) I/O points                    | 1            | 2               | -             | -          |
| <b>PC45.</b> Ensuring zero-material damage while handling the equipment during installation process | 1            | 2               | -             | -          |
| <b>PC46.</b> Performing software functionality test                                                 | 1            | 2               | -             | -          |
| <i>Achieving Quality and Productivity as per company norms</i>                                      | <b>6</b>     | <b>12</b>       | -             | -          |
| <b>PC47.</b> Achieving 100% work schedule as planned for the week                                   | 1            | 2               | -             | -          |
| <b>PC48.</b> Following standard operating procedure of tools and equipment and avoid any hazard     | 1            | 2               | -             | -          |
| <b>PC49.</b> Achieving zero component damage                                                        | 1            | 2               | -             | -          |
| <b>PC50.</b> Keeping work area clean and organized                                                  | 1            | 2               | -             | -          |
| <b>PC51.</b> Achieving 100% compliance with health and safety guidelines and rules                  | 1            | 2               | -             | -          |
| <b>PC52.</b> Ensuring installed components as per company norms and standards                       | 1            | 2               | -             | -          |
| <b>NOS Total</b>                                                                                    | <b>55</b>    | <b>95</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                     |
|----------------------------|---------------------------------------------------------------------|
| <b>NOS Code</b>            | IAS/N2103                                                           |
| <b>NOS Name</b>            | Install and Provide Technical Support for CCTV Surveillance Systems |
| <b>Sector</b>              | Instrumentation                                                     |
| <b>Sub-Sector</b>          | Instrumentation & Automation                                        |
| <b>Occupation</b>          | Product Engineering/System Design                                   |
| <b>NSQF Level</b>          | 5                                                                   |
| <b>Credits</b>             | 3                                                                   |
| <b>Version</b>             | 4.0                                                                 |
| <b>Last Reviewed Date</b>  | 07/10/2025                                                          |
| <b>Next Review Date</b>    | 07/10/2030                                                          |
| <b>NSQC Clearance Date</b> | 07/10/2025                                                          |

## Qualification Pack

# IAS/N2104: Integrating and Controlling Building automation System

## Description

This OS unit is about Integrating CCTV Surveillance Systems, Fire Alarm Systems, Access Control Devices and HVAC Components in the control Panel

## Scope

The scope covers the following :

- This unit/task covers the following: Integrating HVAC Components Integrating Fire Alarm Systems Integrating Access Control Devices Integrating CCTV Surveillance Systems Controlling and Supervising Building Automation Systems using Control Panel.

## Elements and Performance Criteria

### *Integrating HVAC Components*

To be competent, the user/individual on the job must be able to:

- PC1.** Installing and Controlling HVAC Components using DDC Controllers
- PC2.** Graphically monitor, control, alarm and diagnose Building Equipment remotely
- PC3.** Creating communication between DDC Controllers using data Bus
- PC4.** Using BACnet, LON(Echelon) and MODBUS to communicate on data Bus
- PC5.** Integrating installed HVAC Components with other Building Automation Systems using Software and programming on Single Control Panel

### *Integrating Fire Alarm Systems*

To be competent, the user/individual on the job must be able to:

- PC6.** Integrating different detectors such as Heat, Smoke, Flame Ionization Detectors, Beam Detectors etc. with control panel
- PC7.** Using Conventional Systems and analogue addressable systems for fire panels
- PC8.** Creating and testing communication between control panel and detectors
- PC9.** Using Intelligent addressable systems as per the requirement
- PC10.** Integrating Fire Alarm Components with central fire alarm system
- PC11.** Integrating Fire alarm system with centralized control panel

### *Integrating Access Control Devices*

To be competent, the user/individual on the job must be able to:

- PC12.** Inter facing between different networks used in Access Controls systems
- PC13.** Integrating tailgate detectors to remove unauthorized access
- PC14.** Controlling and monitoring multiple doors using reader controllers
- PC15.** Creating communication between control panel and access control servers
- PC16.** Integrating DIU (Door interface Units)
- PC17.** Integrating access control and intrusion detection
- PC18.** Installing Biometric systems on application device

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- PC19.** Installing and creating communication between magnetic locks and doors
- PC20.** Managing Smart card management systems
- PC21.** Integrating Access Control System with Time/Attendance payroll system
- PC22.** Managing Weigand Communication for specific interface between card and readers.
- PC23.** Integrating Access Control devices with BAS control Panel

### *Integrating CCTV Surveillance Systems*

To be competent, the user/individual on the job must be able to:

- PC24.** Managing Iris and Auto Iris functionality of installed CCTV System
- PC25.** Managing Automatic Shutter Speed
- PC26.** Creating communication between CCTV Camera and DVR or NVR
- PC27.** Managing Automatic Gain Control
- PC28.** Managing Synchronization between installed CCTV Camera
- PC29.** Creating communication between IP cameras and network
- PC30.** Managing NVR and NVR Software
- PC31.** Installing and managing Facial and number plate recognition system
- PC32.** Integrating CCTV Surveillance System with Security system to provide centralized management of access control
- PC33.** Integrating centralized Access Control System with Building Automation System control Panel

### *Controlling and Supervising Building Automation Systems using Control Panel*

To be competent, the user/individual on the job must be able to:

- PC34.** Testing of overall integrated Building Automation System through control panel
- PC35.** Ensuring proper working and controlling of every installed device using control panel
- PC36.** Assuring 100% satisfaction from the customer after installation of BAS
- PC37.** Troubleshooting errors if the system is not working as per the requirements

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Company manufacturing processes
- KU2.** Existing layout for the processes
- KU3.** Sequence of operations for each process
- KU4.** Facility planning methodology being followed in the company
- KU5.** Future capacity expansions plans (if any) of the company
- KU6.** Complete knowledge of the BAS Software Integration
- KU7.** Understanding of the latest technologies used in HVAC Systems
- KU8.** Fire alarm system technology and components used
- KU9.** Access controller installation and integration techniques
- KU10.** Standard Electrical/electronic parts and assemblies
- KU11.** Standard Mechanical, pneumatic and hydraulic parts and assemblies
- KU12.** Understanding wiring and drawings for communication

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- KU13.** Integrating all the components as per customers requirement
- KU14.** Understanding refrigerants, chillers and components used in air conditioning system
- KU15.** Understanding fixed air volume and variable air volume applications
- KU16.** Understanding Psychometrics
- KU17.** Knowledge of sensors, actuators, input devices, output devices, logic statements and PID loops
- KU18.** Understanding point scheduling and I/O summary
- KU19.** Knowledge of Building Management System (BMS) graphics screens and tags
- KU20.** Knowledge of Ethernet and I/P & Industrial Networks such as MODBUS, LON, BAC net etc.
- KU21.** Understanding Direct Digital Controllers in BAS
- KU22.** Understanding Laws of Thermodynamics
- KU23.** Understanding characteristics of fire classes
- KU24.** Understanding Fire Detection technologies
- KU25.** Knowledge of Fire panel Technologies
- KU26.** Understanding detectors and device wiring schema
- KU27.** Knowledge of Firemans Telephony, Talkback systems and Mass Evacuation
- KU28.** Understanding Peer-to-peer & Daisy Chain Networks
- KU29.** Understanding NFPA Guidelines and fire safety strategies
- KU30.** Knowledge of Access control System components such as Magnetic swipe, RF proximity cards, smart cards, Mifare smart cards, i-class smart cards, readers etc.
- KU31.** Understanding data encryption and security
- KU32.** Knowledge of complete CCTV Surveillance system and components used
- KU33.** Understanding power requirements of different components
- KU34.** Knowledge of different cables used with their specifications
- KU35.** Knowledge of different types of camera technologies used with their specifications

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Compile all the data related to main and auxiliary equipment required in the process
- GS2.** Compile all the data related to study of existing facility in the form of presentation and reports
- GS3.** Record Faults in instruments.
- GS4.** Read the equipment literature and understand its features
- GS5.** Read the information displayed at the workplace
- GS6.** Discuss task lists, schedules, and work-loads with co-workers
- GS7.** Communicate the new equipment and facilities design activities to the higher management in meetings for their support
- GS8.** Spell out effectively the findings of the study to the higher management in meetings
- GS9.** Interact with coworkers and gather all the information related to process requirement
- GS10.** Make decisions pertaining to the concerned area of work



## Qualification Pack

- GS11.** Plan the execution of entire design activity; long term and short term activities so that he can finish the task activity wise in the stipulated time.
- GS12.** Organize expert support from/within outside professional environment for new technology and feasibility studies.
- GS13.** Understand real needs of the customer and suggest most appropriate solution
- GS14.** Support customers when they need help
- GS15.** Think through the problem, evaluate the possible solution(s) and suggest an optimum /best possible solution(s)
- GS16.** Identify immediate or temporary solutions to resolve delays
- GS17.** Use the existing information to arrive at actionable decision points
- GS18.** Use the existing information for improving the customer satisfaction
- GS19.** Use the existing information to optimize solution and company business
- GS20.** Analyze problems and identify causes and possible solutions
- GS21.** Apply, analyze, and evaluate the information gathered from observation, experience, reasoning, or communication, as a guide to thought and action.
- GS22.** Analyze the way in which the existing facility layout is in operation and think of more economic and feasible measures for existing layout modification.

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Integrating HVAC Components</i>                                                                                                              | <b>9</b>     | <b>10</b>       | -             | -          |
| <b>PC1.</b> Installing and Controlling HVAC Components using DDC Controllers                                                                    | 2            | 2               | -             | -          |
| <b>PC2.</b> Graphically monitor, control, alarm and diagnose Building Equipment remotely                                                        | 2            | 2               | -             | -          |
| <b>PC3.</b> Creating communication between DDC Controllers using data Bus                                                                       | 1            | 2               | -             | -          |
| <b>PC4.</b> Using BACnet, LON(Echelon) and MODBUS to communicate on data Bus                                                                    | 2            | 2               | -             | -          |
| <b>PC5.</b> Integrating installed HVAC Components with other Building Automation Systems using Software and programming on Single Control Panel | 2            | 2               | -             | -          |
| <i>Integrating Fire Alarm Systems</i>                                                                                                           | <b>6</b>     | <b>10</b>       | -             | -          |
| <b>PC6.</b> Integrating different detectors such as Heat, Smoke, Flame Ionization Detectors, Beam Detectors etc. with control panel             | 1            | 2               | -             | -          |
| <b>PC7.</b> Using Conventional Systems and analogue addressable systems for fire panels                                                         | 1            | 2               | -             | -          |
| <b>PC8.</b> Creating and testing communication between control panel and detectors                                                              | 1            | 2               | -             | -          |
| <b>PC9.</b> Using Intelligent addressable systems as per the requirement                                                                        | 1            | 1               | -             | -          |
| <b>PC10.</b> Integrating Fire Alarm Components with central fire alarm system                                                                   | 1            | 1               | -             | -          |
| <b>PC11.</b> Integrating Fire alarm system with centralized control panel                                                                       | 1            | 2               | -             | -          |
| <i>Integrating Access Control Devices</i>                                                                                                       | <b>15</b>    | <b>25</b>       | -             | -          |
| <b>PC12.</b> Inter facing between different networks used in Access Controls systems                                                            | 1            | 2               | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                             | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC13.</b> Integrating tailgate detectors to remove unauthorized access                    | 1            | 1               | -             | -          |
| <b>PC14.</b> Controlling and monitoring multiple doors using reader controllers              | 1            | 1               | -             | -          |
| <b>PC15.</b> Creating communication between control panel and access control servers         | 2            | 1               | -             | -          |
| <b>PC16.</b> Integrating DIU (Door interface Units)                                          | 1            | 3               | -             | -          |
| <b>PC17.</b> Integrating access control and intrusion detection                              | 2            | 4               | -             | -          |
| <b>PC18.</b> Installing Biometric systems on application device                              | 2            | 3               | -             | -          |
| <b>PC19.</b> Installing and creating communication between magnetic locks and doors          | 1            | 3               | -             | -          |
| <b>PC20.</b> Managing Smart card management systems                                          | 1            | 2               | -             | -          |
| <b>PC21.</b> Integrating Access Control System with Time/Attendance payroll system           | 1            | 1               | -             | -          |
| <b>PC22.</b> Managing Weigand Communication for specific interface between card and readers. | 1            | 1               | -             | -          |
| <b>PC23.</b> Integrating Access Control devices with BAS control Panel                       | 1            | 3               | -             | -          |
| <i>Integrating CCTV Surveillance Systems</i>                                                 | <b>12</b>    | <b>26</b>       | -             | -          |
| <b>PC24.</b> Managing Iris and Auto Iris functionality of installed CCTV System              | 2            | 3               | -             | -          |
| <b>PC25.</b> Managing Automatic Shutter Speed                                                | 2            | 4               | -             | -          |
| <b>PC26.</b> Creating communication between CCTV Camera and DVR or NVR                       | 1            | 3               | -             | -          |
| <b>PC27.</b> Managing Automatic Gain Control                                                 | 1            | 2               | -             | -          |
| <b>PC28.</b> Managing Synchronization between installed CCTV Camera                          | 1            | 1               | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                           | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC29.</b> Creating communication between IP cameras and network                                                         | 1            | 3               | -             | -          |
| <b>PC30.</b> Managing NVR and NVR Software                                                                                 | 1            | 3               | -             | -          |
| <b>PC31.</b> Installing and managing Facial and number plate recognition system                                            | 1            | 2               | -             | -          |
| <b>PC32.</b> Integrating CCTV Surveillance System with Security system to provide centralized management of access control | 1            | 3               | -             | -          |
| <b>PC33.</b> Integrating centralized Access Control System with Building Automation System control Panel                   | 1            | 2               | -             | -          |
| <i>Controlling and Supervising Building Automation Systems using Control Panel</i>                                         | <b>4</b>     | <b>8</b>        | -             | -          |
| <b>PC34.</b> Testing of overall integrated Building Automation System through control panel                                | 1            | 2               | -             | -          |
| <b>PC35.</b> Ensuring proper working and controlling of every installed device using control panel                         | 1            | 2               | -             | -          |
| <b>PC36.</b> Assuring 100% satisfaction from the customer after installation of BAS                                        | 1            | 2               | -             | -          |
| <b>PC37.</b> Troubleshooting errors if the system is not working as per the requirements                                   | 1            | 2               | -             | -          |
| <b>NOS Total</b>                                                                                                           | <b>46</b>    | <b>79</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                        |
|----------------------------|--------------------------------------------------------|
| <b>NOS Code</b>            | IAS/N2104                                              |
| <b>NOS Name</b>            | Integrating and Controlling Building automation System |
| <b>Sector</b>              | Instrumentation                                        |
| <b>Sub-Sector</b>          | Instrumentation & Automation                           |
| <b>Occupation</b>          | Product Engineering/System Design                      |
| <b>NSQF Level</b>          | 5                                                      |
| <b>Credits</b>             | 2                                                      |
| <b>Version</b>             | 3.0                                                    |
| <b>Last Reviewed Date</b>  | 07/10/2025                                             |
| <b>Next Review Date</b>    | 07/10/2030                                             |
| <b>NSQC Clearance Date</b> | 07/10/2025                                             |

## Qualification Pack

### DGT/VSQ/N0102: Employability Skills (60 Hours)

#### Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

#### Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

#### Elements and Performance Criteria

##### *Introduction to Employability Skills*

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

##### *Constitutional values – Citizenship*

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

##### *Becoming a Professional in the 21st Century*

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

##### *Basic English Skills*

To be competent, the user/individual on the job must be able to:

## Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

### *Career Development & Goal Setting*

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

### *Communication Skills*

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

### *Diversity & Inclusion*

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

### *Financial and Legal Literacy*

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

### *Essential Digital Skills*

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

### *Entrepreneurship*

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

### *Customer Service*

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

## Qualification Pack

**PC28.** follow appropriate hygiene and grooming standards

*Getting ready for apprenticeship & Jobs*

To be competent, the user/individual on the job must be able to:

**PC29.** create a professional Curriculum vitae (Résumé)

**PC30.** search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

**PC31.** apply to identified job openings using offline /online methods as per requirement

**PC32.** answer questions politely, with clarity and confidence, during recruitment and selection

**PC33.** identify apprenticeship opportunities and register for it as per guidelines and requirements

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** need for employability skills and different learning and employability related portals

**KU2.** various constitutional and personal values

**KU3.** different environmentally sustainable practices and their importance

**KU4.** Twenty first (21st) century skills and their importance

**KU5.** how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

**KU6.** importance of career development and setting long- and short-term goals

**KU7.** about effective communication

**KU8.** POSH Act

**KU9.** Gender sensitivity and inclusivity

**KU10.** different types of financial institutes, products, and services

**KU11.** how to compute income and expenditure

**KU12.** importance of maintaining safety and security in offline and online financial transactions

**KU13.** different legal rights and laws

**KU14.** different types of digital devices and the procedure to operate them safely and securely

**KU15.** how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

**KU16.** how to identify business opportunities

**KU17.** types and needs of customers

**KU18.** how to apply for a job and prepare for an interview

**KU19.** apprenticeship scheme and the process of registering on apprenticeship portal

## Generic Skills (GS)

User/individual on the job needs to know how to:

**GS1.** read and write different types of documents/instructions/correspondence

**GS2.** communicate effectively using appropriate language in formal and informal settings



## Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Introduction to Employability Skills</i>                                                                                                                                                                                                                                                                      | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC1.</b> identify employability skills required for jobs in various industries                                                                                                                                                                                                                                | -            | -               | -             | -          |
| <b>PC2.</b> identify and explore learning and employability portals                                                                                                                                                                                                                                              | -            | -               | -             | -          |
| <i>Constitutional values – Citizenship</i>                                                                                                                                                                                                                                                                       | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC3.</b> recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.                                                               | -            | -               | -             | -          |
| <b>PC4.</b> follow environmentally sustainable practices                                                                                                                                                                                                                                                         | -            | -               | -             | -          |
| <i>Becoming a Professional in the 21st Century</i>                                                                                                                                                                                                                                                               | <b>2</b>     | <b>4</b>        | -             | -          |
| <b>PC5.</b> recognize the significance of 21st Century Skills for employment                                                                                                                                                                                                                                     | -            | -               | -             | -          |
| <b>PC6.</b> practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life | -            | -               | -             | -          |
| <i>Basic English Skills</i>                                                                                                                                                                                                                                                                                      | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC7.</b> use basic English for everyday conversation in different contexts, in person and over the telephone                                                                                                                                                                                                  | -            | -               | -             | -          |
| <b>PC8.</b> read and understand routine information, notes, instructions, mails, letters etc. written in English                                                                                                                                                                                                 | -            | -               | -             | -          |
| <b>PC9.</b> write short messages, notes, letters, e-mails etc. in English                                                                                                                                                                                                                                        | -            | -               | -             | -          |
| <i>Career Development &amp; Goal Setting</i>                                                                                                                                                                                                                                                                     | <b>1</b>     | <b>2</b>        | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                      | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC10.</b> understand the difference between job and career                                                         | -            | -               | -             | -          |
| <b>PC11.</b> prepare a career development plan with short- and long-term goals, based on aptitude                     | -            | -               | -             | -          |
| <i>Communication Skills</i>                                                                                           | <b>2</b>     | <b>2</b>        | -             | -          |
| <b>PC12.</b> follow verbal and non-verbal communication etiquette and active listening techniques in various settings | -            | -               | -             | -          |
| <b>PC13.</b> work collaboratively with others in a team                                                               | -            | -               | -             | -          |
| <i>Diversity &amp; Inclusion</i>                                                                                      | <b>1</b>     | <b>2</b>        | -             | -          |
| <b>PC14.</b> communicate and behave appropriately with all genders and PwD                                            | -            | -               | -             | -          |
| <b>PC15.</b> escalate any issues related to sexual harassment at workplace according to POSH Act                      | -            | -               | -             | -          |
| <i>Financial and Legal Literacy</i>                                                                                   | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC16.</b> select financial institutions, products and services as per requirement                                  | -            | -               | -             | -          |
| <b>PC17.</b> carry out offline and online financial transactions, safely and securely                                 | -            | -               | -             | -          |
| <b>PC18.</b> identify common components of salary and compute income, expenses, taxes, investments etc                | -            | -               | -             | -          |
| <b>PC19.</b> identify relevant rights and laws and use legal aids to fight against legal exploitation                 | -            | -               | -             | -          |
| <i>Essential Digital Skills</i>                                                                                       | <b>3</b>     | <b>4</b>        | -             | -          |
| <b>PC20.</b> operate digital devices and carry out basic internet operations securely and safely                      | -            | -               | -             | -          |
| <b>PC21.</b> use e- mail and social media platforms and virtual collaboration tools to work effectively               | -            | -               | -             | -          |
| <b>PC22.</b> use basic features of word processor, spreadsheets, and presentations                                    | -            | -               | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Entrepreneurship</i>                                                                                                                                                          | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC23.</b> identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research                                       | -            | -               | -             | -          |
| <b>PC24.</b> develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion                                                      | -            | -               | -             | -          |
| <b>PC25.</b> identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity                                           | -            | -               | -             | -          |
| <i>Customer Service</i>                                                                                                                                                          | <b>1</b>     | <b>2</b>        | -             | -          |
| <b>PC26.</b> identify different types of customers                                                                                                                               | -            | -               | -             | -          |
| <b>PC27.</b> identify and respond to customer requests and needs in a professional manner.                                                                                       | -            | -               | -             | -          |
| <b>PC28.</b> follow appropriate hygiene and grooming standards                                                                                                                   | -            | -               | -             | -          |
| <i>Getting ready for apprenticeship &amp; Jobs</i>                                                                                                                               | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC29.</b> create a professional Curriculum vitae (Résumé)                                                                                                                     | -            | -               | -             | -          |
| <b>PC30.</b> search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively | -            | -               | -             | -          |
| <b>PC31.</b> apply to identified job openings using offline /online methods as per requirement                                                                                   | -            | -               | -             | -          |
| <b>PC32.</b> answer questions politely, with clarity and confidence, during recruitment and selection                                                                            | -            | -               | -             | -          |
| <b>PC33.</b> identify apprenticeship opportunities and register for it as per guidelines and requirements                                                                        | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                 | <b>20</b>    | <b>30</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                 |
|----------------------------|---------------------------------|
| <b>NOS Code</b>            | DGT/VSQ/N0102                   |
| <b>NOS Name</b>            | Employability Skills (60 Hours) |
| <b>Sector</b>              | Cross Sectoral                  |
| <b>Sub-Sector</b>          | Professional Skills             |
| <b>Occupation</b>          | Employability                   |
| <b>NSQF Level</b>          | 4                               |
| <b>Credits</b>             | 2                               |
| <b>Version</b>             | 1.0                             |
| <b>Last Reviewed Date</b>  | 18/02/2025                      |
| <b>Next Review Date</b>    | 18/02/2028                      |
| <b>NSQC Clearance Date</b> | 18/02/2025                      |

## Qualification Pack

# IAS/N8014: Design, install, & Provide Technical Support for the HVAC System

## Description

This OS unit is about Design, Installation, Supervision, Testing and providing Technical Support of HVAC Systems involved in building automation systems.

## Scope

The scope covers the following :

- This unit/task covers the following: Capturing the requirements of HVAC Systems by site survey  
Suggesting and taking approval from the customer for HVAC Systems  
Installing approved HVAC components as per site requirements  
Wiring Electrical and Electronics components as per specifications  
Testing of HVAC systems  
Providing Technical Support for HVAC Systems

## Elements and Performance Criteria

### *Capturing the requirements of HVAC Systems by site survey*

To be competent, the user/individual on the job must be able to:

- PC1.** Capturing work requirements of the client by site survey
- PC2.** Developing BOQ according to the requirement of the client
- PC3.** Creating 2D models using BAS Software
- PC4.** Developing program on BAS Tools for HVAC
- PC5.** Developing program on BAS Tools based on Logic gates
- PC6.** Developing program related to Air Conditioning on BAS Tools
- PC7.** Managing wiring of components in AC Drives and Soft Starters

### *Suggesting and taking approval from the customer for HVAC Systems*

To be competent, the user/individual on the job must be able to:

- PC8.** Suggesting appropriate HVAC components to the customer according to the site
- PC9.** Assisting the customer in choosing different types of technologies and specifications used in HVAC Systems
- PC10.** Taking approval from the customer
- PC11.** Maintaining complete documentation of the components to be installed

### *Installing approved HVAC components as per site requirements*

To be competent, the user/individual on the job must be able to:

- PC12.** Collecting and checking of components at customer premises as per checklist
- PC13.** Installing HVAC components including VFD at site as per the requirement
- PC14.** Installing AHU (Air Handling Unit)
- PC15.** Installing Chiller
- PC16.** Installing Sensors
- PC17.** Installing VAV (Variable Air Volume), TFA (Treated Fresh Air) etc.

## Qualification Pack

### *Wiring Electrical and Electronics components as per the requirements*

To be competent, the user/individual on the job must be able to:

- PC18.** Wiring Power Supplies, Earthing & Grounding
- PC19.** Wiring and connecting Shielded & Unshielded Cables, Cable Gauges & AWG sizes
- PC20.** Wiring of HVAC hardware PC

### *Testing of HVAC systems*

To be competent, the user/individual on the job must be able to:

- PC21.** Testing of installed HVAC System
- PC22.** Ensuring proper working of the installed HVAC System
- PC23.** Assuring 100% satisfaction from the customer after installation
- PC24.** Troubleshooting errors if the system is not working as per the requirements

### *Providing Technical Support for HVAC Systems*

To be competent, the user/individual on the job must be able to:

- PC25.** Calculating total number of HVAC controllers as per I/O summary
- PC26.** Managing refrigeration process needed for the site by BAS controller
- PC27.** Managing Wiring and drawings of Components used in Centralized Air-Conditioning
- PC28.** Providing Technical Support for HVAC functions using BAS controller
- PC29.** Calculating Heat Load

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Company policies on delivery standards, services and warranty
- KU2.** Company sales and after sales policies
- KU3.** Sequence of reporting for each process
- KU4.** Companies policies on incentives and reimbursement
- KU5.** Latest technologies introduced by the company for Fire Alarm Systems
- KU6.** Complete knowledge of the HVAC processing consideration
- KU7.** Operation of machinery and equipment being used for HVAC process
- KU8.** Engineering drawings of existing layout/equipment
- KU9.** Electrical/electronic standard parts
- KU10.** Understanding Basic concepts of Electrical and Electronics Engineering
- KU11.** Knowledge of different detection technologies used in fire alarm systems
- KU12.** Material and information flow of HVAC process
- KU13.** Understanding Installation procedures given in the manuals
- KU14.** Power Requirements by different components used in HVAC
- KU15.** Knowledge of safety measures and quality standards of fire alarm components
- KU16.** Knowledge of integrating other systems with fire alarm systems
- KU17.** Compete knowledge of fire alarm system works under considerations

## Qualification Pack

### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Compile all the data related to main and auxiliary equipment required in the process
- GS2.** Compile all the data related to study of existing facility in the form of presentation and reports
- GS3.** Record Faults in instruments.
- GS4.** Read the equipment literature and understand its features
- GS5.** Read the information displayed at the workplace
- GS6.** Discuss task lists, schedules, and work-loads with co-workers
- GS7.** Communicate the new equipment and facilities design activities to the higher management in meetings for their support
- GS8.** Spell out effectively the findings of the study to the higher management in meetings
- GS9.** Interact with coworkers and gather all the information related to process requirement
- GS10.** Make decisions pertaining to the concerned area of work
- GS11.** Plan the execution of entire project detailing long term and short term activities so that he/she can finish the project in the stipulated time.
- GS12.** Organize expert support from/within outside organization for new technology and feasibility studies.
- GS13.** Understand real needs of the customer and suggest most appropriate solution
- GS14.** Support customer when they need help
- GS15.** Think through the problem, evaluate the possible solution(s) and suggest an optimum/best possible solution(s)
- GS16.** Identify immediate or temporary solutions to resolve delays
- GS17.** Use the existing information to arrive at actionable decision points
- GS18.** Use the existing information for improving the customer satisfaction
- GS19.** Use the existing information to optimize solution and company business
- GS20.** Analyze problems and identify causes and possible solutions
- GS21.** Apply, analyze, and evaluate the information gathered from observation, experience, reasoning, or communication, as a guide to thought and action
- GS22.** Analyze the way in which the existing facility layout is in operation and think of more economic and feasible measures for existing layout modification.



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                       | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Capturing the requirements of HVAC Systems by site survey</i>                                                       | <b>10</b>    | <b>12</b>       | -             | -          |
| <b>PC1.</b> Capturing work requirements of the client by site survey                                                   | 1            | 2               | -             | -          |
| <b>PC2.</b> Developing BOQ according to the requirement of the client                                                  | 1            | 2               | -             | -          |
| <b>PC3.</b> Creating 2D models using BAS Software                                                                      | 2            | 2               | -             | -          |
| <b>PC4.</b> Developing program on BAS Tools for HVAC                                                                   | 1            | 2               | -             | -          |
| <b>PC5.</b> Developing program on BAS Tools based on Logic gates                                                       | 2            | 2               | -             | -          |
| <b>PC6.</b> Developing program related to Air Conditioning on BAS Tools                                                | 1            | 1               | -             | -          |
| <b>PC7.</b> Managing wiring of components in AC Drives and Soft Starters                                               | 2            | 1               | -             | -          |
| <i>Suggesting and taking approval from the customer for HVAC Systems</i>                                               | <b>6</b>     | <b>10</b>       | -             | -          |
| <b>PC8.</b> Suggesting appropriate HVAC components to the customer according to the site                               | 2            | 3               | -             | -          |
| <b>PC9.</b> Assisting the customer in choosing different types of technologies and specifications used in HVAC Systems | 2            | 2               | -             | -          |
| <b>PC10.</b> Taking approval from the customer                                                                         | 1            | 2               | -             | -          |
| <b>PC11.</b> Maintaining complete documentation of the components to be installed                                      | 1            | 3               | -             | -          |
| <i>Installing approved HVAC components as per site requirements</i>                                                    | <b>4</b>     | <b>4</b>        | -             | -          |
| <b>PC12.</b> Collecting and checking of components at customer premises as per checklist                               | 1            | -               | -             | -          |
| <b>PC13.</b> Installing HVAC components including VFD at site as per the requirement                                   | 1            | 1               | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                             | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC14.</b> Installing AHU (Air Handling Unit)                                              | 1            | 1               | -             | -          |
| <b>PC15.</b> Installing Chiller                                                              | -            | 1               | -             | -          |
| <b>PC16.</b> Installing Sensors                                                              | -            | 1               | -             | -          |
| <b>PC17.</b> Installing VAV (Variable Air Volume), TFA (Treated Fresh Air) etc.              | 1            | -               | -             | -          |
| <i>Wiring Electrical and Electronics components as per the requirements</i>                  | <b>8</b>     | <b>13</b>       | -             | -          |
| <b>PC18.</b> Wiring Power Supplies, Earthing & Grounding                                     | 4            | 4               | -             | -          |
| <b>PC19.</b> Wiring and connecting Shielded & Unshielded Cables, Cable Gauges & AWG sizes    | 2            | 4               | -             | -          |
| <b>PC20.</b> Wiring of HVAC hardware PC                                                      | 2            | 5               | -             | -          |
| <i>Testing of HVAC systems</i>                                                               | <b>8</b>     | <b>13</b>       | -             | -          |
| <b>PC21.</b> Testing of installed HVAC System                                                | 3            | 5               | -             | -          |
| <b>PC22.</b> Ensuring proper working of the installed HVAC System                            | 2            | 3               | -             | -          |
| <b>PC23.</b> Assuring 100% satisfaction from the customer after installation                 | 2            | 2               | -             | -          |
| <b>PC24.</b> Troubleshooting errors if the system is not working as per the requirements     | 1            | 3               | -             | -          |
| <i>Providing Technical Support for HVAC Systems</i>                                          | <b>4</b>     | <b>8</b>        | -             | -          |
| <b>PC25.</b> Calculating total number of HVAC controllers as per I/O summary                 | 1            | 1               | -             | -          |
| <b>PC26.</b> Managing refrigeration process needed for the site by BAS controller            | -            | 1               | -             | -          |
| <b>PC27.</b> Managing Wiring and drawings of Components used in Centralized Air-Conditioning | -            | 1               | -             | -          |
| <b>PC28.</b> Providing Technical Support for HVAC functions using BAS controller             | -            | 3               | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------|--------------|-----------------|---------------|------------|
| PC29. Calculating Heat Load      | 3            | 2               | -             | -          |
| <b>NOS Total</b>                 | <b>40</b>    | <b>60</b>       | <b>-</b>      | <b>-</b>   |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                  |
|----------------------------|------------------------------------------------------------------|
| <b>NOS Code</b>            | IAS/N8014                                                        |
| <b>NOS Name</b>            | Design, install, & Provide Technical Support for the HVAC System |
| <b>Sector</b>              | Instrumentation                                                  |
| <b>Sub-Sector</b>          | Instrumentation & Automation                                     |
| <b>Occupation</b>          | Product Engineering/System Design                                |
| <b>NSQF Level</b>          | 5                                                                |
| <b>Credits</b>             | 4                                                                |
| <b>Version</b>             | 1.0                                                              |
| <b>Last Reviewed Date</b>  | 07/10/2025                                                       |
| <b>Next Review Date</b>    | 07/10/2030                                                       |
| <b>NSQF Clearance Date</b> | 07/10/2025                                                       |

## Qualification Pack

# IAS/N8015: Design Install and Provide Technical Support for Fire Alarm System

## Description

This OS unit is about design, installation, supervision and testing of Fire Alarm Systems in building automation systems.

## Scope

The scope covers the following :

- This unit/task covers the following:
- Capturing the requirements of Fire Alarm Systems by site survey
- Suggesting and taking approval from the customer for Fire Alarm System
- Installing approved Fire Alarm components as per site requirements
- Wiring Electrical and Electronics components as per specifications Testing of new systems at customer premises
- Providing Technical Support for Fire Alarm Systems at the site

## Elements and Performance Criteria

### *Capturing the requirements of Fire Alarm Systems by site survey*

To be competent, the user/individual on the job must be able to:

- PC1.** Capturing work requirements of the client by site survey
- PC2.** Developing BOQ according to the requirement of the client
- PC3.** Assisting customer about different types of technologies used in FAS according to the need of site
- PC4.** Ensuring that Suggesting components matches to customers requirement
- PC5.** Assisting the customers about the company policies towards services and warranty
- PC6.** Managing proper documentation of site survey and customers requirements

### *Suggesting and taking approval from the customer for Fire Alarm Systems*

To be competent, the user/individual on the job must be able to:

- PC7.** Suggesting appropriate FAS components to the customer according to the site
- PC8.** Assisting the customers about Fire Alarm systems with their specifications
- PC9.** Taking approval for installing FAS components from the customer
- PC10.** Preparing and assembling FAS components as per the requirement
- PC11.** Creating check list before going to the site for installation
- PC12.** Maintaining complete documentation of the components to be installed

### *Installing approved Fire Alarm components as per site requirements*

To be competent, the user/individual on the job must be able to:

- PC13.** Collecting and checking of components before moving to customer premises
- PC14.** Assisting technicians for checking hardware components before FAS Installation

## Qualification Pack

- PC15.** Replacing FAS components if found malfunctioning
- PC16.** Preparing Checklist and ensure the availability of every component before installation
- PC17.** Installing FAS components at the customers site
- PC18.** Installing fire Detection components as per site requirement
- PC19.** Installing Heat/RoR Detectors, Smoke Detectors and Multi Criteria detectors as per requirement

### *Wiring Electrical and Electronics components as per the requirements*

To be competent, the user/individual on the job must be able to:

- PC20.** Wiring Power Supplies, Earthing & Grounding
- PC21.** Wiring and connecting Shielded & Unshielded Cables, Cable Gauges, SWG & AWG sizes
- PC22.** Ensuring adequate length of connecting cables as per the requirements
- PC23.** Wiring of FAS hardware
- PC24.** Using proper terminals as prescribed for joining cables
- PC25.** Use power cable for connecting power supply with proper rating

### *Testing of new systems at customer site*

To be competent, the user/individual on the job must be able to:

- PC26.** Testing installed FAS components at customer premises
- PC27.** Ensuring proper working of every component
- PC28.** Ensuring proper working of FAS systems before leaving the site and explaining the customer how to operate the system properly
- PC29.** Assuring 100% satisfaction from the customer after installation
- PC30.** Troubleshooting the errors if the system is not working as per the requirements

### *Providing Technical Support for Fire Alarm Systems at the site*

To be competent, the user/individual on the job must be able to:

- PC31.** Providing Technical Support for Fire Detection & Alarm System as per Fire Life cycle and Class of Fire
- PC32.** Providing Technical Support for intelligent Fire Panels & conventional Fire Panels installed
- PC33.** Managing Detector & Device Wiring Schema
- PC34.** Integrating Fireman's Telephony & Talkback system with fire alarm
- PC35.** Selecting Fire Safety Strategies for prevention

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Company policies on delivery standards, services and warranty
- KU2.** Company sales and after sales policies
- KU3.** Sequence of reporting for each process
- KU4.** Companies policies on incentives and reimbursement
- KU5.** Latest technologies introduced by the company for Fire Alarm Systems
- KU6.** Complete knowledge of the FAS process in consideration
- KU7.** Operation of machinery and equipment being used for FAS process

## Qualification Pack

- KU8.** Engineering drawings of existing layout/equipment
- KU9.** Electrical/electronic standard parts
- KU10.** Understanding Basic concepts of Electrical and Electronics Engineering
- KU11.** Knowledge of different detection technologies used in fire alarm systems
- KU12.** Material and information flow of FAS process
- KU13.** Understanding Installation procedures given in the manuals
- KU14.** Power Requirements by different components used in FAS
- KU15.** Knowledge of safety measures and quality standards of fire alarm components
- KU16.** Knowledge of integrating other systems with fire alarm systems
- KU17.** Compete knowledge of fire alarm system works under considerations

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Compile all the data related to main and auxiliary equipment required in the process
- GS2.** Compile all the data related to study of existing facility in the form of presentation and reports
- GS3.** Record Faults in instruments.
- GS4.** Read the equipment literature and understand its features
- GS5.** Read the information displayed at the workplace
- GS6.** Discuss task lists, schedules, and work-loads with co-workers
- GS7.** Communicate the new equipment and facilities required to the higher management in meetings for their support
- GS8.** Spell out effectively the findings of the study to the higher management in meetings
- GS9.** Interact with coworkers and gather all the information related to process requirement
- GS10.** Make decisions pertaining to the concerned area of work
- GS11.** Plan the execution of entire project detailing long term and short term activities so that he/she can finish the project in the stipulated time.
- GS12.** Organize expert support from/within outside organization for new technology and feasibility studies.
- GS13.** Understand real needs of the customer and suggest most appropriate solution
- GS14.** Support customer when they need help
- GS15.** Think through the problem, evaluate the possible solution(s) and suggest an optimum /best possible solution(s)
- GS16.** Identify immediate or temporary solutions to resolve delays
- GS17.** Use the existing information to arrive at actionable decision points
- GS18.** Use the existing information for improving the customer satisfaction
- GS19.** Use the existing information to optimize solution and company business
- GS20.** Analyze problems and identify causes and possible solutions
- GS21.** Apply, analyze, and evaluate the information gathered from observation, experience, reasoning, or communication, as a guide to thought and action

## Qualification Pack

- GS22.** Analyze the way in which the existing facility layout is in operation and think of more economic and feasible measures for existing layout modification.



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                               | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Capturing the requirements of Fire Alarm Systems by site survey</i>                                         | <b>10</b>    | <b>12</b>       | -             | -          |
| <b>PC1.</b> Capturing work requirements of the client by site survey                                           | 1            | 2               | -             | -          |
| <b>PC2.</b> Developing BOQ according to the requirement of the client                                          | 2            | 2               | -             | -          |
| <b>PC3.</b> Assisting customer about different types of technologies used in FAS according to the need of site | 2            | 2               | -             | -          |
| <b>PC4.</b> Ensuring that Suggesting components matches to customers requirement                               | 2            | 2               | -             | -          |
| <b>PC5.</b> Assisting the customers about the company policies towards services and warranty                   | 2            | 2               | -             | -          |
| <b>PC6.</b> Managing proper documentation of site survey and customers requirements                            | 1            | 2               | -             | -          |
| <i>Suggesting and taking approval from the customer for Fire Alarm Systems</i>                                 | <b>6</b>     | <b>10</b>       | -             | -          |
| <b>PC7.</b> Suggesting appropriate FAS components to the customer according to the site                        | 1            | 2               | -             | -          |
| <b>PC8.</b> Assisting the customers about Fire Alarm systems with their specifications                         | 1            | 3               | -             | -          |
| <b>PC9.</b> Taking approval for installing FAS components from the customer                                    | 1            | 2               | -             | -          |
| <b>PC10.</b> Preparing and assembling FAS components as per the requirement                                    | 1            | 1               | -             | -          |
| <b>PC11.</b> Creating check list before going to the site for installation                                     | -            | -               | -             | -          |
| <b>PC12.</b> Maintaining complete documentation of the components to be installed                              | 2            | 2               | -             | -          |
| <i>Installing approved Fire Alarm components as per site requirements</i>                                      | <b>5</b>     | <b>8</b>        | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                            | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC13.</b> Collecting and checking of components before moving to customer premises                       | 1            | 1               | -             | -          |
| <b>PC14.</b> Assisting technicians for checking hardware components before FAS Installation                 | -            | 1               | -             | -          |
| <b>PC15.</b> Replacing FAS components if found malfunctioning                                               | 1            | 1               | -             | -          |
| <b>PC16.</b> Preparing Checklist and ensure the availability of every component before installation         | 1            | 1               | -             | -          |
| <b>PC17.</b> Installing FAS components at the customers site                                                | 1            | 1               | -             | -          |
| <b>PC18.</b> Installing fire Detection components as per site requirement                                   | 1            | 1               | -             | -          |
| <b>PC19.</b> Installing Heat/RoR Detectors, Smoke Detectors and Multi Criteria detectors as per requirement | -            | 2               | -             | -          |
| <i>Wiring Electrical and Electronics components as per the requirements</i>                                 | <b>8</b>     | <b>13</b>       | -             | -          |
| <b>PC20.</b> Wiring Power Supplies, Earthing & Grounding                                                    | 1            | 2               | -             | -          |
| <b>PC21.</b> Wiring and connecting Shielded & Unshielded Cables, Cable Gauges, SWG & AWG sizes              | 2            | 2               | -             | -          |
| <b>PC22.</b> Ensuring adequate length of connecting cables as per the requirements                          | 1            | 2               | -             | -          |
| <b>PC23.</b> Wiring of FAS hardware                                                                         | 1            | 3               | -             | -          |
| <b>PC24.</b> Using proper terminals as prescribed for joining cables                                        | 1            | 2               | -             | -          |
| <b>PC25.</b> Use power cable for connecting power supply with proper rating                                 | 2            | 2               | -             | -          |
| <i>Testing of new systems at customer site</i>                                                              | <b>8</b>     | <b>13</b>       | -             | -          |
| <b>PC26.</b> Testing installed FAS components at customer premises                                          | 2            | 1               | -             | -          |
| <b>PC27.</b> Ensuring proper working of every component                                                     | 1            | 3               | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                                           | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC28.</b> Ensuring proper working of FAS systems before leaving the site and explaining the customer how to operate the system properly | 2            | 2               | -             | -          |
| <b>PC29.</b> Assuring 100% satisfaction from the customer after installation                                                               | 1            | 3               | -             | -          |
| <b>PC30.</b> Troubleshooting the errors if the system is not working as per the requirements                                               | 2            | 4               | -             | -          |
| <i>Providing Technical Support for Fire Alarm Systems at the site</i>                                                                      | <b>3</b>     | <b>4</b>        | -             | -          |
| <b>PC31.</b> Providing Technical Support for Fire Detection & Alarm System as per Fire Life cycle and Class of Fire                        | 1            | 1               | -             | -          |
| <b>PC32.</b> Providing Technical Support for intelligent Fire Panels & conventional Fire Panels installed                                  | -            | 1               | -             | -          |
| <b>PC33.</b> Managing Detector & Device Wiring Schema                                                                                      | 1            | -               | -             | -          |
| <b>PC34.</b> Integrating Fireman's Telephony & Talkback system with fire alarm                                                             | 1            | -               | -             | -          |
| <b>PC35.</b> Selecting Fire Safety Strategies for prevention                                                                               | -            | 2               | -             | -          |
| <b>NOS Total</b>                                                                                                                           | <b>40</b>    | <b>60</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                    |
|----------------------------|--------------------------------------------------------------------|
| <b>NOS Code</b>            | IAS/N8015                                                          |
| <b>NOS Name</b>            | Design Install and Provide Technical Support for Fire Alarm System |
| <b>Sector</b>              | Instrumentation                                                    |
| <b>Sub-Sector</b>          | Instrumentation & Automation                                       |
| <b>Occupation</b>          | Product Engineering/System Design                                  |
| <b>NSQF Level</b>          | 5                                                                  |
| <b>Credits</b>             | 4                                                                  |
| <b>Version</b>             | 1.0                                                                |
| <b>Last Reviewed Date</b>  | 07/10/2025                                                         |
| <b>Next Review Date</b>    | 07/10/2030                                                         |
| <b>NSQC Clearance Date</b> | 07/10/2025                                                         |

## Qualification Pack

# IAS/N8016: Install and Provide Technical Support for Access Control Systems

## Description

This OS unit is about Installing and Commissioning Access Controls Systems at the customers premises

## Scope

The scope covers the following :

- This unit/task covers the following: Capturing the requirements of Access Controls Systems by site survey Suggesting and taking approval from the customer for Access Controls System Installing approved Access Controls components as per site requirements Wiring Electrical and Electronics components as per specifications Testing Access Control systems at customer premises Providing Technical Support for Access Controls Systems Achieving Quality and Productivity as per company norms

## Elements and Performance Criteria

### *Capturing the requirements of Access Controls Systems by site survey*

To be competent, the user/individual on the job must be able to:

- PC1.** Capturing work requirements of the client by site survey
- PC2.** Developing BOQ according to the requirement of the client
- PC3.** Assisting customer about different types of technologies used in Access Control Systems according to the need of site
- PC4.** Ensuring that Suggesting components matches to customers requirement
- PC5.** Assisting the customers about the company policies towards services and warranty
- PC6.** Managing proper documentation of site survey and customers requirements

### *Suggesting and taking approval from the customer for Access Controls System*

To be competent, the user/individual on the job must be able to:

- PC7.** Suggesting appropriate Access Controls components to the customer according to the site
- PC8.** Assisting the customers about technologies used in Access Control systems with their specifications
- PC9.** Taking approval for installing Access Control Systems from the customer
- PC10.** Preparing and assembling Access Control Systems components as per the requirement
- PC11.** Creating check list before going to the site for installation
- PC12.** Maintaining complete documentation of the components to be installed

### *Installing approved Access Controls components as per site requirements*

To be competent, the user/individual on the job must be able to:

- PC13.** Collecting and checking of components before moving to customer premises
- PC14.** Assisting technicians for checking hardware components before Installation
- PC15.** Replacing components if found malfunctioning

## Qualification Pack

- PC16.** Preparing Checklist and ensure the availability of every component before installation
- PC17.** Installing components at the customers site
- PC18.** Installing hardware such as smart hub, RFID Card, Door control unit, card readers etc.
- PC19.** Ensuring that components are matching with customers requirement and installed as per standard operating operation

### *Wiring Electrical and Electronics components as per specifications*

To be competent, the user/individual on the job must be able to:

- PC20.** Determining the type of cable requirement for different types of network type such as USB, twisted pair, etc.
- PC21.** Ensuring adequate length of cables are available for installation
- PC22.** Wiring of Power Supplies, Earthing & Grounding.

### *Testing Access Control systems at customer premises*

To be competent, the user/individual on the job must be able to:

- PC23.** Checking voltage and resistance at all appropriate points of the system
- PC24.** Correcting alignment and operation of access point hardware
- PC25.** Verifying access levels
- PC26.** Checking correct operation of each reader
- PC27.** Testing Release time for each lock using software
- PC28.** Checking the signals if doors are held open and signaling is required
- PC29.** Checking all the data for correct entry in the ACS software
- PC30.** Checking alarms to display correctly
- PC31.** Defining level of particular object in the software

### *Providing Technical Support for Access Controls Systems*

To be competent, the user/individual on the job must be able to:

- PC32.** Identifying the operating system and software requirement for the access control system
- PC33.** Providing Technical Support for Access controls devices at the customer premises
- PC34.** Commissioning Access Controls Systems performance as per customer requirements
- PC35.** Achieving zero errors in commissioning as per company policy
- PC36.** Identifying problems and alert on time
- PC37.** Fixing for any errors (if any) identified
- PC38.** Verifying software implementation checks of input/output I/O points (AI, AO, DI,DO)
- PC39.** Verifying sensor calibration, control sequence logic, graphics and alarm code
- PC40.** Performing software functionality test

### *Achieving Quality and Productivity as per company norms*

To be competent, the user/individual on the job must be able to:

- PC41.** Achieving 100% work schedule as planned for the week
- PC42.** Meeting 100% daily or monthly target
- PC43.** Achieving zero component damage
- PC44.** Keeping work area clean and organized
- PC45.** Identifying problems and alert in time
- PC46.** Achieving 100% compliance with health and safety guidelines and rules

## Qualification Pack

### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Company's policies on: incentives, delivery standards and personnel management, customer management
- KU2.** Reporting and documentation processes
- KU3.** Importance of the individual's role in the workflow
- KU4.** Reporting structure
- KU5.** Access Control System process
- KU6.** Operation of components used in Access Control system
- KU7.** Engineering drawings and layout of site
- KU8.** Electrical/electronic/mechanical parts required for the ACS
- KU9.** Material requirement of components with their specifications
- KU10.** Equipment used in Access Controls system
- KU11.** Knowledge of Graphic Generation Tools in BAS
- KU12.** Understanding of Direct Digital Controllers in BAS
- KU13.** Knowledge of communication between ACS and other devices
- KU14.** Knowledge of programming in BAS softwares
- KU15.** Knowledge of devices communication protocols
- KU16.** Knowledge of integrating communication protocols
- KU17.** Knowledge of different types of cables used for data transmission and power transmission
- KU18.** Power requirements of hardware
- KU19.** Different types of access controls hardware available in the market

### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Preparing indents and list of equipment required for the specified Access Controls System
- GS2.** Compiling all the related documents as per the requirements
- GS3.** Record Faults in instruments and submit to the team leader.
- GS4.** Read warnings, instructions and other text material on product levels, components etc.
- GS5.** Reading drawings and job sheets or work orders
- GS6.** Discuss work load with superiors
- GS7.** Communicate with team leader for proper understanding of the work
- GS8.** Interact with coworkers and gather all the information related to process requirement
- GS9.** Make decisions according to the situations
- GS10.** Plan the execution of entire design activity; long term and short term activities so that he can finish the task activity wise in the stipulated time.
- GS11.** Organize expert support from/within outside professional environment for new technology and feasibility studies.

## Qualification Pack

- GS12.** Understand real needs of the customer and suggest appropriate solution
- GS13.** Support customers when they need help
- GS14.** Think and provide best possible solution required for the system
- GS15.** Identify immediate or temporary solutions to resolve delays
- GS16.** Use the existing information to arrive at actionable decision points
- GS17.** Use the existing information for improving the customer satisfaction
- GS18.** Use the existing information to optimize solution and company business
- GS19.** Analyze problems and identify causes and possible solutions
- GS20.** Analyze the work flow and suggest improvements to provide improvement for better production



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                  | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Capturing the requirements of Access Controls Systems by site survey</i>                                                       | <b>9</b>     | <b>9</b>        | -             | -          |
| <b>PC1.</b> Capturing work requirements of the client by site survey                                                              | 2            | 1               | -             | -          |
| <b>PC2.</b> Developing BOQ according to the requirement of the client                                                             | 2            | 2               | -             | -          |
| <b>PC3.</b> Assisting customer about different types of technologies used in Access Control Systems according to the need of site | 1            | 2               | -             | -          |
| <b>PC4.</b> Ensuring that Suggesting components matches to customers requirement                                                  | 2            | 1               | -             | -          |
| <b>PC5.</b> Assisting the customers about the company policies towards services and warranty                                      | 1            | 1               | -             | -          |
| <b>PC6.</b> Managing proper documentation of site survey and customers requirements                                               | 1            | 2               | -             | -          |
| <i>Suggesting and taking approval from the customer for Access Controls System</i>                                                | <b>6</b>     | <b>8</b>        | -             | -          |
| <b>PC7.</b> Suggesting appropriate Access Controls components to the customer according to the site                               | 1            | 2               | -             | -          |
| <b>PC8.</b> Assisting the customers about technologies used in Access Control systems with their specifications                   | 1            | 1               | -             | -          |
| <b>PC9.</b> Taking approval for installing Access Control Systems from the customer                                               | 1            | 1               | -             | -          |
| <b>PC10.</b> Preparing and assembling Access Control Systems components as per the requirement                                    | 1            | 1               | -             | -          |
| <b>PC11.</b> Creating check list before going to the site for installation                                                        | 1            | 2               | -             | -          |
| <b>PC12.</b> Maintaining complete documentation of the components to be installed                                                 | 1            | 1               | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                                | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Installing approved Access Controls components as per site requirements</i>                                                  | <b>9</b>     | <b>17</b>       | -             | -          |
| <b>PC13.</b> Collecting and checking of components before moving to customer premises                                           | 1            | 1               | -             | -          |
| <b>PC14.</b> Assisting technicians for checking hardware components before Installation                                         | 1            | 1               | -             | -          |
| <b>PC15.</b> Replacing components if found malfunctioning                                                                       | 1            | 3               | -             | -          |
| <b>PC16.</b> Preparing Checklist and ensure the availability of every component before installation                             | 2            | 4               | -             | -          |
| <b>PC17.</b> Installing components at the customers site                                                                        | 2            | 3               | -             | -          |
| <b>PC18.</b> Installing hardware such as smart hub, RFID Card, Door control unit, card readers etc.                             | 1            | 3               | -             | -          |
| <b>PC19.</b> Ensuring that components are matching with customers requirement and installed as per standard operating operation | 1            | 2               | -             | -          |
| <i>Wiring Electrical and Electronics components as per specifications</i>                                                       | <b>3</b>     | <b>6</b>        | -             | -          |
| <b>PC20.</b> Determining the type of cable requirement for different types of network type such as USB, twisted pair, etc.      | 1            | 1               | -             | -          |
| <b>PC21.</b> Ensuring adequate length of cables are available for installation                                                  | 1            | 2               | -             | -          |
| <b>PC22.</b> Wiring of Power Supplies, Earthing & Grounding.                                                                    | 1            | 3               | -             | -          |
| <i>Testing Access Control systems at customer premises</i>                                                                      | <b>4</b>     | <b>10</b>       | -             | -          |
| <b>PC23.</b> Checking voltage and resistance at all appropriate points of the system                                            | -            | 1               | -             | -          |
| <b>PC24.</b> Correcting alignment and operation of access point hardware                                                        | 1            | 1               | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                     | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC25.</b> Verifying access levels                                                                 | -            | 1               | -             | -          |
| <b>PC26.</b> Checking correct operation of each reader                                               | 1            | 1               | -             | -          |
| <b>PC27.</b> Testing Release time for each lock using software                                       | -            | 1               | -             | -          |
| <b>PC28.</b> Checking the signals if doors are held open and signaling is required                   | 1            | -               | -             | -          |
| <b>PC29.</b> Checking all the data for correct entry in the ACS software                             | -            | 1               | -             | -          |
| <b>PC30.</b> Checking alarms to display correctly                                                    | 1            | 2               | -             | -          |
| <b>PC31.</b> Defining level of particular object in the software                                     | -            | 2               | -             | -          |
| <i>Providing Technical Support for Access Controls Systems</i>                                       | <b>4</b>     | <b>5</b>        | -             | -          |
| <b>PC32.</b> Identifying the operating system and software requirement for the access control system | 1            | -               | -             | -          |
| <b>PC33.</b> Providing Technical Support for Access controls devices at the customer premises        | -            | 1               | -             | -          |
| <b>PC34.</b> Commissioning Access Controls Systems performance as per customer requirements          | 1            | -               | -             | -          |
| <b>PC35.</b> Achieving zero errors in commissioning as per company policy                            | -            | 1               | -             | -          |
| <b>PC36.</b> Identifying problems and alert on time                                                  | 1            | -               | -             | -          |
| <b>PC37.</b> Fixing for any errors (if any) identified                                               | -            | 1               | -             | -          |
| <b>PC38.</b> Verifying software implementation checks of input/output I/O points (AI, AO, DI,DO)     | 1            | -               | -             | -          |
| <b>PC39.</b> Verifying sensor calibration, control sequence logic, graphics and alarm code           | -            | 1               | -             | -          |
| <b>PC40.</b> Performing software functionality test                                                  | -            | 1               | -             | -          |
| <i>Achieving Quality and Productivity as per company norms</i>                                       | <b>5</b>     | <b>5</b>        | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                   | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC41.</b> Achieving 100% work schedule as planned for the week                  | 1            | -               | -             | -          |
| <b>PC42.</b> Meeting 100% daily or monthly target                                  | -            | 1               | -             | -          |
| <b>PC43.</b> Achieving zero component damage                                       | 1            | -               | -             | -          |
| <b>PC44.</b> Keeping work area clean and organized                                 | -            | 1               | -             | -          |
| <b>PC45.</b> Identifying problems and alert in time                                | 2            | 1               | -             | -          |
| <b>PC46.</b> Achieving 100% compliance with health and safety guidelines and rules | 1            | 2               | -             | -          |
| <b>NOS Total</b>                                                                   | <b>40</b>    | <b>60</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                  |
|----------------------------|------------------------------------------------------------------|
| <b>NOS Code</b>            | IAS/N8016                                                        |
| <b>NOS Name</b>            | Install and Provide Technical Support for Access Control Systems |
| <b>Sector</b>              | Instrumentation                                                  |
| <b>Sub-Sector</b>          | Instrumentation & Automation                                     |
| <b>Occupation</b>          | Product Engineering/System Design                                |
| <b>NSQF Level</b>          | 5                                                                |
| <b>Credits</b>             | 3                                                                |
| <b>Version</b>             | 1.0                                                              |
| <b>Last Reviewed Date</b>  | 07/10/2025                                                       |
| <b>Next Review Date</b>    | 07/10/2030                                                       |
| <b>NSQC Clearance Date</b> | 07/10/2025                                                       |

## Qualification Pack

# IAS/N8012: Integration of Smart Technologies in HVAC, Lighting, Security and other building Automation

## Description

This NOS unit is about to Integrate Smart Technologies in HVAC, lighting, security, and other building automation systems.

## Scope

The scope covers the following :

- Assessing the specific operational requirements of Smart Technologies.
- Developing integrated platforms that centralize control and monitoring of all building systems.
- Deploying advanced monitoring systems that provide real-time insights.

## Elements and Performance Criteria

### *Assessing the specific operational requirements of Smart Technologies*

To be competent, the user/individual on the job must be able to:

- PC1.** Evaluate the ability of new smart technologies to integrate seamlessly with existing building management systems (BMS).
- PC2.** Assess the intuitiveness and user-friendliness of the system interfaces for operators and occupants.
- PC3.** Check the system's capability for real-time analytics and reporting features.
- PC4.** Examine the ability of the system to scale up or down based on changing operational needs or technological advancements.
- PC5.** Analyze the total cost of ownership, including upfront costs, ongoing maintenance, and operational savings.

### *Developing integrated platforms that centralize control and monitoring of all building systems*

To be competent, the user/individual on the job must be able to:

- PC6.** Develop a user-friendly interface that provides real-time monitoring and control of all building systems.
- PC7.** Implement a central control unit that can process inputs from various building systems and execute commands accordingly.
- PC8.** Define user roles and permissions to limit access to sensitive data and functions.
- PC9.** Incorporate tools for tracking energy usage and generating reports to support energy efficiency initiatives.
- PC10.** Regularly assess cybersecurity measures and aim for zero breaches or significant incidents.

### *Deploying advanced monitoring systems that provide real-time insights.*

To be competent, the user/individual on the job must be able to:

- PC11.** Enhance the Capability to manage all building systems (HVAC, lighting, security) from a single interface.
- PC12.** Implement systems to analyze and optimize energy consumption across all building systems.

## Qualification Pack

- PC13.** Measure the percentage of devices successfully integrated with the monitoring system.
- PC14.** Track the rate of successfully identified vs. undetected anomalies in system performance.
- PC15.** Evaluate the system's ability to incorporate future advancements in smart technologies.
- PC16.** Develop Automated alerts for users regarding system performance anomalies or failures.
- PC17.** Identify and connect all relevant data sources, including sensors, controllers, and monitoring systems.
- PC18.** Utilize a centralized database to securely store collected data for easy retrieval and analysis.
- PC19.** Develop automated report generation capabilities that can produce performance reports on a scheduled basis or upon request.
- PC20.** Create user-friendly dashboards that visualize system performance data in real-time, with options for drill-downs into detailed reports.

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understanding the key components of HVAC systems (e.g., chillers, boilers, ductwork, and control systems) and their operational dynamics
- KU2.** Knowledge of variable refrigerant flow (VRF), smart thermostats, and energy recovery ventilators (ERVs).
- KU3.** Familiarity with various control methods, such as direct digital control (DDC), modulating control, and on/off control.
- KU4.** Understanding different types of lighting (LED, fluorescent, etc.) and smart technologies (e.g., occupancy sensors, daylight harvesting).
- KU5.** Knowledge of lighting control protocols such as DALI (Digital Addressable Lighting Interface) or Zigbee.
- KU6.** Familiarity with how lighting systems integrate with building automation systems (BAS) for dynamic controls and scheduled operations.
- KU7.** Knowledge of technologies like RFID, biometric scanners, and keyless entry systems
- KU8.** Understanding of IP cameras, motion detectors, and advanced surveillance monitoring systems.
- KU9.** Familiarity with intrusion detection systems and alarm management software.
- KU10.** Understanding of BACnet, Modbus, LonWorks, and MQTT communication protocols for data exchange between devices.

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicating effectively in a global organization
- GS2.** collaborating extensively within the organization to achieve common goals
- GS3.** collaborating with customers to understand their needs and translating that into solution specifications
- GS4.** collaborating with partners and vendors to achieve the organization goals

## Qualification Pack

**GS5.** negotiation skills with customers, vendors, across disciplines and roles



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                     | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Assessing the specific operational requirements of Smart Technologies</i>                                                         | 5            | 20              | -             | -          |
| <b>PC1.</b> Evaluate the ability of new smart technologies to integrate seamlessly with existing building management systems (BMS).  | 1            | 4               | -             | -          |
| <b>PC2.</b> Assess the intuitiveness and user-friendliness of the system interfaces for operators and occupants.                     | 1            | 4               | -             | -          |
| <b>PC3.</b> Check the system's capability for real-time analytics and reporting features.                                            | 1            | 4               | -             | -          |
| <b>PC4.</b> Examine the ability of the system to scale up or down based on changing operational needs or technological advancements. | 1            | 4               | -             | -          |
| <b>PC5.</b> Analyze the total cost of ownership, including upfront costs, ongoing maintenance, and operational savings.              | 1            | 4               | -             | -          |
| <i>Developing integrated platforms that centralize control and monitoring of all building systems</i>                                | 5            | 20              | -             | -          |
| <b>PC6.</b> Develop a user-friendly interface that provides real-time monitoring and control of all building systems.                | 1            | 4               | -             | -          |
| <b>PC7.</b> Implement a central control unit that can process inputs from various building systems and execute commands accordingly. | 1            | 4               | -             | -          |
| <b>PC8.</b> Define user roles and permissions to limit access to sensitive data and functions.                                       | 1            | 4               | -             | -          |
| <b>PC9.</b> Incorporate tools for tracking energy usage and generating reports to support energy efficiency initiatives.             | 1            | 4               | -             | -          |
| <b>PC10.</b> Regularly assess cybersecurity measures and aim for zero breaches or significant incidents.                             | 1            | 4               | -             | -          |
| <i>Deploying advanced monitoring systems that provide real-time insights.</i>                                                        | 20           | 30              | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                      | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC11.</b> Enhance the Capability to manage all building systems (HVAC, lighting, security) from a single interface.                                | 2            | 3               | -             | -          |
| <b>PC12.</b> Implement systems to analyze and optimize energy consumption across all building systems.                                                | 2            | 3               | -             | -          |
| <b>PC13.</b> Measure the percentage of devices successfully integrated with the monitoring system.                                                    | 2            | 3               | -             | -          |
| <b>PC14.</b> Track the rate of successfully identified vs. undetected anomalies in system performance.                                                | 2            | 3               | -             | -          |
| <b>PC15.</b> Evaluate the system's ability to incorporate future advancements in smart technologies.                                                  | 2            | 3               | -             | -          |
| <b>PC16.</b> Develop Automated alerts for users regarding system performance anomalies or failures.                                                   | 2            | 3               | -             | -          |
| <b>PC17.</b> Identify and connect all relevant data sources, including sensors, controllers, and monitoring systems.                                  | 2            | 3               | -             | -          |
| <b>PC18.</b> Utilize a centralized database to securely store collected data for easy retrieval and analysis.                                         | 2            | 3               | -             | -          |
| <b>PC19.</b> Develop automated report generation capabilities that can produce performance reports on a scheduled basis or upon request.              | 2            | 3               | -             | -          |
| <b>PC20.</b> Create user-friendly dashboards that visualize system performance data in real-time, with options for drill-downs into detailed reports. | 2            | 3               | -             | -          |
| <b>NOS Total</b>                                                                                                                                      | <b>30</b>    | <b>70</b>       | <b>-</b>      | <b>-</b>   |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                                             |
|----------------------------|---------------------------------------------------------------------------------------------|
| <b>NOS Code</b>            | IAS/N8012                                                                                   |
| <b>NOS Name</b>            | Integration of Smart Technologies in HVAC, Lighting, Security and other building Automation |
| <b>Sector</b>              | Instrumentation                                                                             |
| <b>Sub-Sector</b>          |                                                                                             |
| <b>Occupation</b>          | Product Engineering/System Design                                                           |
| <b>NSQF Level</b>          | 5.0                                                                                         |
| <b>Credits</b>             | 1                                                                                           |
| <b>Version</b>             | 1.0                                                                                         |
| <b>Last Reviewed Date</b>  | 07/10/2025                                                                                  |
| <b>Next Review Date</b>    | 07/10/2030                                                                                  |
| <b>NSQC Clearance Date</b> | 07/10/2025                                                                                  |

## Assessment Guidelines and Assessment Weightage

### Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.
6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack

## Qualification Pack

**Minimum Aggregate Passing % at QP Level : 70**

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

## Assessment Weightage

Compulsory NOS

| National Occupational Standards                                                                      | Theory Marks | Practical Marks | Project Marks | Viva Marks | Total Marks | Weightage  |
|------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|-------------|------------|
| IAS/N2103.Install and Provide Technical Support for CCTV Surveillance Systems                        | 55           | 95              | -             | -          | 150         | 20         |
| IAS/N2104.Integrating and Controlling Building automation System                                     | 46           | 79              | -             | -          | 125         | 20         |
| DGT/VSQ/N0102.Employability Skills (60 Hours)                                                        | 20           | 30              | 0             | 0          | 50          | 10         |
| IAS/N8014.Design, install, & Provide Technical Support for the HVAC System                           | 40           | 60              | -             | -          | 100         | 20         |
| IAS/N8015.Design Install and Provide Technical Support for Fire Alarm System                         | 40           | 60              | -             | -          | 100         | 10         |
| IAS/N8016.Install and Provide Technical Support for Access Control Systems                           | 40           | 60              | -             | -          | 100         | 10         |
| IAS/N8012.Integration of Smart Technologies in HVAC,Lighting, Security and other building Automation | 30           | 70              | -             | -          | 100         | 10         |
| <b>Total</b>                                                                                         | <b>271</b>   | <b>454</b>      | <b>-</b>      | <b>-</b>   | <b>725</b>  | <b>100</b> |

## Qualification Pack

### Acronyms

|             |                                                 |
|-------------|-------------------------------------------------|
| <b>NOS</b>  | National Occupational Standard(s)               |
| <b>NSQF</b> | National Skills Qualifications Framework        |
| <b>QP</b>   | Qualifications Pack                             |
| <b>TVET</b> | Technical and Vocational Education and Training |

## Qualification Pack

### Glossary

|                                              |                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sector</b>                                | Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.                                                                      |
| <b>Sub-sector</b>                            | Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.                                                                                                                                                                                     |
| <b>Occupation</b>                            | Occupation is a set of job roles, which perform similar/ related set of functions in an industry.                                                                                                                                                                                                |
| <b>Job role</b>                              | Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.                                                                                                                                                                                |
| <b>Occupational Standards (OS)</b>           | OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts. |
| <b>Performance Criteria (PC)</b>             | Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.                                                                                                                                                                    |
| <b>National Occupational Standards (NOS)</b> | NOS are occupational standards which apply uniquely in the Indian context.                                                                                                                                                                                                                       |
| <b>Qualifications Pack (QP)</b>              | QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.                                                                                                                       |
| <b>Unit Code</b>                             | Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'                                                                                                                                                                                                        |
| <b>Unit Title</b>                            | Unit title gives a clear overall statement about what the incumbent should be able to do.                                                                                                                                                                                                        |
| <b>Description</b>                           | Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.                                                                                                                   |
| <b>Scope</b>                                 | Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.                                                                                            |

## Qualification Pack

|                                         |                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge and Understanding (KU)</b> | Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.                                                                                                                                               |
| <b>Organisational Context</b>           | Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.                                                                                                                                                                       |
| <b>Technical Knowledge</b>              | Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.                                                                                                                                                                                                                                                               |
| <b>Core Skills/ Generic Skills (GS)</b> | Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles. |
| <b>Electives</b>                        | Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.                                                                            |
| <b>Options</b>                          | Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.                                                                                                                                                          |